

St. Mary's University



Computer Science Department

Course title:- Programing Fundamentals ||

(GROUP ASSIGHMENT) : - PHARMACY MANEGMENT SYSTEM

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Introduction

1. Using by class

The **Pharmacy Inventory Management System** is a console-based application developed in C++ to automate and organize pharmacy operations. The system replaces manual record-keeping with a digital approach that improves accuracy, efficiency, and data security. It is designed to manage medicine records, control inventory levels, and facilitate interaction between pharmacy administrators and customers.

The application is built using **Object-Oriented Programming (OOP)** principles to ensure modularity, maintainability, and clear separation of responsibilities. By applying concepts such as **encapsulation, constructors, and file handling**, the system simulates a real-world pharmacy environment while remaining simple and easy to use through a text-based interface.

System Overview

The application consists of four main components: Medicine, Admin, User, and Main Page. Each component is implemented using classes and interacts through menu-driven control flow. The system ensures data persistence using files such as MyMedicines.txt, Mystock.txt, MyCart.txt, and Report.txt.

Medicine Class

The Medicine class :- is the core of the system. It stores medicine attributes such as ID, name, price, and quantity. Encapsulation is applied by keeping data members private and accessing them through getter and setter methods. The class also handles file formatting and parsing using stringstream for safe storage and retrieval.

File Handling Mechanism

Medicine records are stored in text files in a comma-separated format. Each object is converted into a single line using a formatting function and reconstructed using a parsing function. Temporary files are used for delete and update operations to ensure data integrity.

Inventory Management

Admins can add, remove, view, and search medicines in the inventory. Stock quantities can be increased or decreased, and updated records are saved back to files. The admin also has access to customer reports submitted through the system. User Operations Customers can register, log in, view available medicines, purchase items, search stock, view cart contents, and submit reports. All customer actions interact with the Medicine class and stored files.

Report System

Customers submit reports that are stored in a text file. Admins can view these reports using word-by-word file streaming to ensure complete output display. Program Flow The program starts at the main page, where users select Admin or User mode. Each mode

Conclusion

The Pharmacy Inventory Management System demonstrates the practical use of **C++ OOP concepts** in solving real-world problems. By transitioning from struct-based design to a class-based architecture, the system achieves better data protection, cleaner organization, and improved maintainability. It serves as an effective educational project that combines programming fundamentals with real-life applications

Using by Struct :-

Introduction

This program is a **console-based Pharmacy Inventory Management System** developed using C++. The main purpose of the system is to computerize and simplify pharmacy operations such as managing medicines, handling stock, and allowing customers to purchase medicines through a menu-driven interface.

The system is designed using **structured programming concepts** with the help of **structs**, **functions**, and **global data structures**. It separates responsibilities between **administrators** and **users**, allowing each role to interact with the system according to its permissions. The application also makes use of **file handling** to store and retrieve medicine and stock data, ensuring that information is not lost when the program terminates.

This project simulates real-world pharmacy operations and serves as a practical learning example for understanding arrays, file handling, menus, and data management in C++.

System Description

Overall Structure

The program starts from a **main menu (mainpage)** that allows the user to choose between:

- Admin mode
- User mode
- Exit

Based on the selected option, the system redirects the user to the appropriate module.

Data Structures Used

The program uses multiple **structs** to organize data:

- **Medicine_Inventory**
Stores all medicines added by the admin, including ID, name, price, quantity, and total length.
- **Medicine_Stock**
Stores medicines that are available for sale to users. It is populated from the inventory.

- **Cart**
Temporarily stores medicines selected by users for purchase.
- **Admin**
Contains all administrator-related operations such as adding medicines, deleting records, sorting prices, managing stock, saving data to files, and viewing customer messages.
- **Users**
Handles user-related operations such as viewing medicines, purchasing, searching, sorting, cart management, and reporting issues.

Admin Module Description

The **Admin Module** provides full control over the pharmacy system. Its main features include:

- Adding medicines to the inventory
- Deleting medicines from inventory
- Viewing and searching inventory medicines
- Adding medicines from inventory into stock
- Removing medicines from stock
- Sorting medicines by price (low to high and high to low)
- Saving inventory data to a file (`Medicine_name.txt`)
- Reading inventory data from a file
- Viewing customer-reported issues
- Changing admin username and password
- Switching to user mode

This module ensures that only authorized personnel can manage critical pharmacy data.

User Module Description

The **User Module** is designed for customers and provides a simple and interactive experience. Users can:

- View available medicines in stock
- Search for medicines by ID
- Purchase medicines and add them to a cart
- View cart contents

- Sort medicines by price
- Save stock data to a file (`Medicine_in_Stock.txt`)
- Read stock data from a file
- Report issues or complaints to the admin
- Switch to admin mode

This module focuses on usability and simulates a real pharmacy shopping process.

File Handling and Data Persistence

The program uses **fstream** to store and retrieve data from text files. This allows:

- Inventory data to persist after program termination
- Stock information to be reused
- Admins and users to access previously saved records

Files are written in a simple, readable text format for easy inspection.

Control Flow and Navigation

The system uses:

- Menu-driven navigation
- Switch-case statements
- Loop-based processing
- Helper functions like `clearScreen()` and continuation functions to improve user interaction

This makes the program interactive and easy to navigate.

Conclusion

This Pharmacy Inventory Management System demonstrates the effective use of **C++ programming fundamentals**, including structs, arrays, file handling, functions, and menu-driven design. It successfully models a real-world pharmacy scenario by separating admin and user responsibilities and maintaining persistent data storage. The project is well-suited for academic learning and provides a strong foundation for future improvements such as database integration or graphical user interfaces

We have attached both sources codes using by **struct** and By **Class** .